

## Linear Algebra

\* 1.

Let  $K$  be a field, and let  $A$  be an  $n \times n$  matrix over  $K$ . Suppose that  $f(x) \in K[x]$  is an irreducible polynomial such that  $f(A) = 0$ . Show that  $\deg(f) \mid n$ .

**Solution 1.**

Let  $m(x) \in K[x]$  be the minimal polynomial of  $A$ . Since  $f(A) = 0$ ,  $m(x)$  must divide  $f(x)$ , but  $f(x)$  is irreducible so  $m(x) = f(x)$ . Let  $\chi(x)$  be the characteristic polynomial of  $A$ , which has degree  $n$ . Since  $\chi(x)$  and  $f(x)$  have the same roots and  $f(x)$  is irreducible, it must be the case that  $\chi(x) = f(x)^\ell$  for some  $\ell$ . Therefore,  $\ell \cdot \deg(f) = \deg(\chi) = n$ , so  $\deg(f) \mid n$ .

2.

Give two examples of matrices in Jordan form with minimal polynomial  $x(x-1)^2(x-2)$  and characteristic polynomial  $x^2(x-1)^4(x-2)$ .

**Solution 2.**

We are looking for  $7 \times 7$  matrices that have eigenvalues 0, 1, and 2 with maximal Jordan blocks of size 1, 2, and 1 respectively. Here are two such matrices:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

3.

Let  $V$  be a finite-dimensional vector space over a field. Let  $A : V \rightarrow V$  be an endomorphism, and suppose  $W \subseteq V$  is an invariant subspace (so that  $A(W) \subseteq W$ ). Let  $\bar{A} : V/W \rightarrow V/W$  be defined by  $\bar{A}(v + W) = Av + W$  for all  $v \in V$ .

(a) Show that  $\bar{A}$  is well-defined.

(b) Show that if  $\bar{A}$  is not invertible, then  $A$  is not invertible.

**Solution 3.**

(a) We need to check that if  $v$  and  $v'$  are two representatives of the same element in  $V/W$ , then  $\bar{A}(v + W) = \bar{A}(v' + W)$ . Since  $v$  and  $v'$  represent the same coset, we have  $v' = v + w$  for some  $w \in W$ . Then by definition, we have

$$\bar{A}(v' + W) = Av' + W = A(v + w) + W = Av + Aw + W = Av + W = \bar{A}(v + W),$$

where the second-to-last equality follows since  $Aw \in W$ .

- (b) We prove the contrapositive. Suppose  $A$  is invertible with inverse  $B$ . We will show that  $\overline{A}$  is invertible by constructing an inverse. First, we show that  $B(W) \subseteq W$ . Since  $A$  is invertible and can be considered as a linear transformation on  $W$ , we must have  $A(W) = W$ . Applying the inverse  $B$  to both sides gives that  $W = B(W)$ . Thus,  $W$  is an invariant subspace of  $B$  as well. Define  $\overline{B} : V/W \rightarrow V/W$  by  $\overline{B}(v + W) = Bv + W$ . As in the previous part, this is well-defined and is indeed an inverse for  $\overline{A}$  since for any  $v + W \in V/W$  we have  $\overline{BA}(v + W) = \overline{B}(Av + W) = BAv + W = v + W$  so  $\overline{BA} = Id$ , and similarly  $\overline{AB} = Id$ .

\* 4.

Let  $V$  be a finite-dimensional vector space over a field. Let  $A : V \rightarrow V$  be a linear transformation.

- (a) Show that there exists  $B : V \rightarrow V$  such that  $ABA = A$ .
- (b) Show that there exists  $C : V \rightarrow V$  such that  $ACA = A$  and  $CAC = C$ .

**Solution 4.**

- (a) This will follow from (b) taking  $B = C$ .
- (b) By using rational canonical form, we may assume that  $A$  is a companion matrix for a polynomial  $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ . First observe that  $A$  is invertible if and only if 0 is not an eigenvalue, which is equivalent to  $x$  not dividing  $p(x)$ , i.e.  $a_0 \neq 0$ . We now split into two cases.

If  $A$  is invertible, we may choose  $C = A^{-1}$ , and then it is clear that  $ACA = A$  and  $CAC = C$ .

If  $A$  is not invertible, then  $a_0 = 0$ , so  $A$  has the form

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & -a_1 \\ 0 & 1 & 0 & \cdots & -a_2 \\ 0 & 0 & 1 & \cdots & -a_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} \end{pmatrix}.$$

Observe that if

$$C = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

then we have

$$AC = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

and one can check that  $ACA = A$  and  $CAC = C$ .

\* 5.

Let  $F$  be a field, and let  $V$  be a finite-dimensional vector space over  $F$  that has dimension  $n \geq 1$ . Let  $q$  be a nonzero element of  $F$  such that  $q^i \neq 1$  for  $1 \leq i \leq n$ . Let  $X : V \rightarrow V$  and  $Y : V \rightarrow V$  be two  $F$ -linear transformations such that  $XY = qYX$ . Show that  $XY$  is nilpotent.

**Solution 5.**

Suppose  $\lambda \neq 0$  is an eigenvalue of  $XY$ . Then  $XYv = \lambda v$  for some nonzero  $v \in V$ , and since  $XY = qYX$ , we also have that  $XY(Yv) = qYXYv = qY(XYv) = q\lambda Yv$ . So,  $q\lambda$  is also an eigenvalue for  $XY$  with eigenvector  $Yv$ . (Note that  $Yv \neq 0$  since  $XYv \neq 0$ .) Iterating this gives that  $q^i\lambda$  is an eigenvalue of  $XY$  for each  $0 \leq i \leq n$ . Since  $q^i \neq 1$  for  $1 \leq i \leq n$ , this means that  $XY$  has at least  $n + 1$  distinct eigenvalues, which is impossible since  $V$  is  $n$ -dimensional. Therefore,  $\lambda = 0$  is the only eigenvalue of  $XY$ , and hence  $XY$  is nilpotent.

6.

Let  $V, W$  be two finite-dimensional vector spaces over  $\mathbb{C}$ . Let  $A : V \rightarrow V$  and  $B : W \rightarrow W$  be linear transformations. List the eigenvalues of  $A \otimes B : V \otimes W \rightarrow V \otimes W$  in terms of the eigenvalues of  $A$  and  $B$ .

**Solution 6.**

We can suppose that  $A$  and  $B$  are in Jordan canonical form. Then  $A$  and  $B$  are upper triangular (with respect to particular bases on  $V$  and  $W$ ) with diagonal elements  $a_{11}, \dots, a_{nn}$  and  $b_{11}, \dots, b_{mm}$  where  $\dim(V) = n$  and  $\dim(W) = m$ . Then  $V \otimes W$  has a basis given by the tensor product of the chosen bases for  $V$  and  $W$ , and  $A \otimes B$  is an upper triangular matrix with diagonal elements  $a_{ii}b_{jj}$  for  $1 \leq i \leq n$  and  $1 \leq j \leq m$ . So, these are the eigenvalues of  $A \otimes B$ .

\* 7.

Let  $X$  be a subspace of  $M_n(\mathbb{C})$ , the  $\mathbb{C}$ -vector space of all  $n \times n$  complex matrices. Assume that every nonzero matrix in  $X$  is invertible. Prove that  $\dim_{\mathbb{C}}(X) \leq 1$ .

**Solution 7.**

Let  $A$  and  $B$  be arbitrary matrices in  $X$ . Since they are both invertible, consider the matrix  $AB^{-1}$ . Since we are over  $\mathbb{C}$  this matrix must have an eigenvalue  $\lambda$ . So,  $AB^{-1} - \lambda \text{Id}$  is not invertible, which means that  $\det(AB^{-1} - \lambda \text{Id}) = 0$ . Then we have

$$\det(A - \lambda B) = \det(AB^{-1} - \lambda \text{Id}) \det(B) = 0$$

which implies that  $A - \lambda B = 0$  since it is an element of  $X$  that is not invertible. Thus,  $A = \lambda B$ . Since all of the matrices in  $X$  are scalar multiples of each other, we have that  $\dim_{\mathbb{C}}(X) \leq 1$ .

\* **8.**

Let  $n > 0$  be an integer. Let  $F$  be a field of characteristic zero,  $V$  a vector space over  $F$  of dimension  $n$ , and  $T : V \rightarrow V$  an invertible  $F$ -linear map such that  $T^{-1} = T$ . Denote by  $W$  the vector space of linear transformations from  $V$  to  $V$  that commute with  $T$ . Find a formula for  $\dim(W)$  in terms of  $n$  and the trace of  $T$ .

**Solution 8.**

Since  $T^{-1} = T$ , we see that  $T^2 - \text{Id}_n = 0$ , so  $T$  satisfies the polynomial  $x^2 - 1 \in F[x]$ . Therefore, the minimal polynomial of  $T$  must divide  $x^2 - 1 = (x - 1)(x + 1)$ . So the only possible eigenvalues of  $T$  are 1 and  $-1$ . Since  $T$  is invertible and its own inverse, we can choose a basis so that  $T$  is a diagonal matrix with only 1's and  $-1$ 's on the diagonal. Furthermore, we can rearrange the basis so that  $T$  has the block matrix form

$$T = \begin{pmatrix} \text{Id}_a & 0 \\ 0 & -\text{Id}_b \end{pmatrix}$$

where  $a + b = n$ . So the trace of  $T$  is  $\text{tr}(T) = a - b$ .

We want to understand matrices that commute with  $T$ . Given an  $n \times n$  matrix  $S$ , we consider it as a block matrix of the form  $S = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$  where  $A$  has size  $a \times a$  and  $D$  has size  $b \times b$ . Then multiplying by  $T$  on either side gives

$$TS = \begin{pmatrix} A & B \\ -C & -D \end{pmatrix} \quad \text{and} \quad ST = \begin{pmatrix} A & -B \\ C & -D \end{pmatrix}.$$

Therefore,  $S \in W$  if and only if  $B = -B = 0$  and  $C = -C = 0$ . Since there are no constraints on  $A$  and  $D$ , we see that  $\dim(W) = a^2 + b^2 = \frac{1}{2}((a + b)^2 + (a - b)^2) = \frac{1}{2}(n^2 + \text{tr}(T)^2)$ .

\* **9.**

Let  $F$  be a field and  $n$  be a positive integer. Fix an  $n \times n$  matrix  $S$  over  $F$  this is invertible and symmetric. Writing  $A^t$  for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F \mid A^t = SAS^{-1}\}.$$

Note that  $V$  is a vector space (you do not need to prove this). Find the dimension of  $V$  in terms of  $n$ .

**Solution 9.**

Since  $S$  is an arbitrary invertible symmetric matrix, let's first think about  $V$  when  $S = \text{Id}_n$ . In this case,  $V$  is the space of matrices that are equal to transpose, i.e. those that are symmetric. Thus,  $\dim(V) = n^2 + (n - 1)^2 + \cdots + 1 = \frac{1}{2}(n^2 + n)$ . Now the question is, why is this the dimension of  $V$  for any invertible symmetric  $S$ ?

Observe that  $A^t = SAS^{-1}$  if and only if  $A^tS = SA$ , and since  $S^t = S$ , this is equivalent to  $(SA)^t = SA$ . So, we want to find the dimension of the space of matrices  $A$  such that  $SA$  is symmetric. Consider the map  $M : M_n(F) \rightarrow M_n(F)$  defined by  $A \mapsto SA$ . Since  $S$  is invertible, this map is an isomorphism. Therefore,  $\dim(V)$  is the dimension of the space of symmetric matrices, which is exactly  $\frac{1}{2}(n^2 + n)$  as before.